

Course 3071

3 Credits: 3

Program Core

Source-grounded

# Chemical Engineering

□ To develop an understanding of thermodynamic systems.

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3071 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3071.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item                 | Official information                        |
|----------------------|---------------------------------------------|
| Programme            | Diploma in Chemical Engineering             |
| Course code          | 3071                                        |
| Course title         | Chemical Engineering                        |
| Semester             | 3 Credits: 3                                |
| Course category      | Program Core                                |
| Periods per week     | 3 (L: 2 T: 1 P: 0) Periods per semester: 45 |
| Periods per semester | 45                                          |

| Item         | Official information                                 |
|--------------|------------------------------------------------------|
| Credits      | 3                                                    |
| Prerequisite | See the official syllabus PDF                        |
| Assessment   | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

20 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- To develop an understanding of thermodynamic systems.
- To enable the students to use thermodynamics concepts in chemical engineering applications.
- To provide an overview of chemical reaction engineering.

### Course outcomes

| CO  | Official outcome                                                                                                           | Study level           |
|-----|----------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | 9 Applying thermodynamics. Interpret the fundamental principles of energy                                                  | See official syllabus |
| CO2 | 16 Applying balance and laws of thermodynamics. Summarize the applications of thermodynamics                               | See official syllabus |
| CO3 | in real world systems and thermodynamic 12 Understanding properties of fluids Outline the fundamental concepts of reaction | See official syllabus |
| CO4 | 8 Understanding engineering.                                                                                               | See official syllabus |

### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3                      |
| CO2 | CO2 3                      |
| CO3 | CO3 2                      |
| CO4 | CO4 2                      |

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

### Module coverage

| Module   | Official heading                                          | Topic entries | Hours        |
|----------|-----------------------------------------------------------|---------------|--------------|
| Module 1 | Interpret the fundamental concepts of thermodynamics.     | 5             | As specified |
| Module 2 | thermodynamics.                                           | 7             | As specified |
| Module 3 | thermodynamic properties of fluids.                       | 4             | As specified |
| Module 4 | Outline the fundamental concepts of reaction engineering. | 4             | As specified |

### MODULE 1

## Interpret the fundamental concepts of thermodynamics.

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Interpret the fundamental concepts of thermodynamics.

- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## Summarize the scope of thermodynamics. 2 Understanding Summarize the properties of a thermodynamic

Summarize the scope of thermodynamics. 2 Understanding Summarize the properties of a thermodynamic is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize the scope of thermodynamics. 2 Understanding Summarize the properties of a thermodynamic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize the scope of thermodynamics. 2 understanding summarize the properties of a thermodynamic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarize the scope of thermodynamics. 2 Understanding Summarize the properties of a thermodynamic means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-താപഗതികശാസ്ത്രത്തിന്റെ പരിധി. 2 Understanding Summarize the properties of a thermodynamic system താപഗതികശാസ്ത്രത്തിന്റെ പരിധി definition/meaning താപഗതികശാസ്ത്രത്തിന്റെ പരിധി. താപഗതികശാസ്ത്രത്തിന്റെ പരിധി, steps, application താപഗതികശാസ്ത്രത്തിന്റെ പരിധി. Technical terms English-താപഗതികശാസ്ത്രത്തിന്റെ പരിധി താപഗതികശാസ്ത്രത്തിന്റെ പരിധി.

## 1 Understanding system.

1 Understanding system. is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 1 Understanding system. means in the official course context                 |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- 1 Understanding system. ഏകദേശം 10000 വാക്കുകളിൽ definition/meaning ഉൾപ്പെടെയുള്ളതായിരിക്കണം. ഉൾപ്പെടെയുള്ളതായിരിക്കണം, steps, application ഉൾപ്പെടെയുള്ളതായിരിക്കണം. Technical terms English-ൽ ഉൾപ്പെടെയുള്ളതായിരിക്കണം.

## Analyse state of a system and phase rule

Analyse state of a system and phase rule is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Analyse state of a system and phase rule using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, analyse state of a system and phase rule should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Definition/identity | What Analyse state of a system and phase rule means in the official course context |
| Study lens          | Principle → components or stages → result → application                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഓൾ ആലൈസ് സ്റ്റേറ്റ് ഓഫ് ഓ സിസ്റ്റം ആൻഡ് ഫേസ് റൂൾ  
ഓൾ ഓൾ ഓൾ ഓൾ definition/meaning ഓൾ ഓൾ ഓൾ ഓൾ. ഓൾ ഓൾ ഓൾ ഓൾ,  
steps, application ഓൾ ഓൾ ഓൾ ഓൾ. Technical terms English-ഓൾ ഓൾ  
ഓൾ ഓൾ ഓൾ.

## Interpret Zeroth law of thermodynamics

Interpret Zeroth law of thermodynamics is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Interpret Zeroth law of thermodynamics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, interpret zeroth law of thermodynamics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Interpret Zeroth law of thermodynamics means in the official course context  |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Interpret Zeroth law of thermodynamics

definition/meaning . steps, application . Technical terms English- .

**Interpret first law of thermodynamics. 4 Applying Scope of thermodynamics - Definitions and fundamental concepts in thermodynamics - System - Process - Surroundings - Homogeneous and Heterogeneous systems - Closed, Open, Isolated and Adiabatic systems. Thermodynamic state and properties - Intensive and Extensive properties-State and Path functions - Heat and work. Steady state - Equilibrium state - Phase rule Temperature and Zeroth law of thermodynamics - Heat reservoir - Heat engine - Heat pump - Reversible and irreversible processes. Statements of first law of thermodynamics - Kinetic energy - Potential energy - Internal energy - Cyclic process- Non flow process - Enthalpy - Simple numerical - Flow process - Heat capacity . Interpret the fundamental principles of energy balance and laws of**

Interpret first law of thermodynamics. 4 Applying Scope of thermodynamics - Definitions and fundamental concepts in thermodynamics - System - Process - Surroundings - Homogeneous and Heterogeneous systems - Closed, Open, Isolated and Adiabatic systems. Thermodynamic state and properties - Intensive and Extensive properties-State and Path functions - Heat and work. Steady state - Equilibrium state - Phase rule Temperature and Zeroth law of thermodynamics - Heat reservoir - Heat engine - Heat pump - Reversible and irreversible processes. Statements of first law of thermodynamics - Kinetic energy - Potential energy - Internal energy - Cyclic process- Non flow process - Enthalpy - Simple numerical - Flow process - Heat capacity . Interpret the fundamental principles of energy balance and laws of is an official topic in Module 1 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Interpret first law of thermodynamics. 4 Applying Scope of thermodynamics - Definitions and fundamental concepts in thermodynamics - System - Process - Surroundings - Homogeneous and Heterogeneous systems - Closed, Open, Isolated and Adiabatic systems. Thermodynamic state and properties - Intensive and Extensive properties-State and Path functions - Heat and work. Steady state - Equilibrium state - Phase rule Temperature and Zeroth law of thermodynamics - Heat reservoir - Heat engine - Heat pump - Reversible and irreversible processes. Statements of first law of thermodynamics - Kinetic energy

- Potential energy - Internal energy - Cyclic process- Non flow process - Enthalpy - Simple numerical - Flow process - Heat capacity . Interpret the fundamental principles of energy balance and laws of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, interpret first law of thermodynamics. 4 applying scope of thermodynamics - definitions and fundamental concepts in thermodynamics - system - process - surroundings - homogeneous and heterogeneous systems - closed, open, isolated and adiabatic systems. thermodynamic state and properties - intensive and extensive properties-state and path functions - heat and work. steady state - equilibrium state - phase rule temperature and zeroth law of thermodynamics - heat reservoir - heat engine - heat pump - reversible and irreversible processes. statements of first law of thermodynamics - kinetic energy - potential energy - internal energy - cyclic process- non flow process - enthalpy - simple numerical - flow process - heat capacity . interpret the fundamental principles of energy balance and laws of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Interpret first law of thermodynamics. 4 Applying Scope of thermodynamics - Definitions and fundamental concepts in thermodynamics - System - Process - Surroundings - Homogeneous and Heterogeneous systems - Closed, Open, Isolated and Adiabatic systems. Thermodynamic state and properties - Intensive and Extensive properties-State and Path functions - Heat and work. Steady state - Equilibrium state - Phase rule Temperature and Zeroth law of thermodynamics - Heat reservoir - Heat engine - Heat pump - Reversible and irreversible processes. Statements of first law of thermodynamics - Kinetic energy - Potential energy - Internal energy - Cyclic process- Non flow process - Enthalpy - Simple numerical - Flow process - Heat capacity . Interpret the fundamental principles of energy balance and laws of means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Interpret first law of thermodynamics. 4  
Applying Scope of thermodynamics - Definitions and fundamental concepts in  
thermodynamics - System - Process - Surroundings - Homogeneous and  
Heterogeneous systems - Closed, Open, Isolated and Adiabatic systems.  
Thermodynamic state and properties - Intensive and Extensive properties-State and  
Path functions - Heat and work. Steady state - Equilibrium state - Phase rule  
Temperature and Zeroth law of thermodynamics - Heat reservoir - Heat engine -  
Heat pump - Reversible and irreversible processes. Statements of first law of  
thermodynamics - Kinetic energy - Potential energy - Internal energy - Cyclic  
process- Non flow process - Enthalpy - Simple numerical - Flow process - Heat  
capacity . Interpret the fundamental principles of energy balance and laws of  
conservation of energy definition/meaning . Interpret the fundamental principles,  
steps, application of energy balance . Technical terms English-Malayalam  
dictionary.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Interpret the fundamental concepts of thermodynamics.

|               |                                             |   |
|---------------|---------------------------------------------|---|
| M1.01         | Summarize the scope of thermodynamics.      | 2 |
| Understanding | Summarize the properties of a thermodynamic |   |
| M1.02         | system.                                     | 1 |
| Understanding |                                             |   |
| M1.03         | Analyse state of a system and phase rule    | 1 |
| Understanding |                                             |   |
| M1.04         | Interpret Zeroth law of thermodynamics      | 1 |
| Understanding |                                             |   |
| M1.05         | Interpret first law of thermodynamics.      | 4 |
| Applying      |                                             |   |

Contents:

Scope of thermodynamics - Definitions and fundamental concepts in thermodynamics -

System – Process - Surroundings - Homogeneous and Heterogeneous systems – Closed, Open, Isolated and Adiabatic systems.

Thermodynamic state and properties - Intensive and Extensive properties-State and Path

functions - Heat and work.

Steady state – Equilibrium state - Phase rule

Temperature and Zeroth law of thermodynamics – Heat reservoir – Heat engine –

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

# thermodynamics.

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: thermodynamics.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## Outline P-V-T behavior of pure fluids

Outline P-V-T behavior of pure fluids is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline P-V-T behavior of pure fluids using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline p-v-t behavior of pure fluids should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Outline P-V-T behavior of pure fluids means in the official course context   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഓൗട്ലൈൻ P-V-T behavior of pure fluids

ഓൗട്ലൈൻ ഓൗട്ലൈൻ definition/meaning ഓൗട്ലൈൻ. ഓൗട്ലൈൻ ഓൗട്ലൈൻ ഓൗട്ലൈൻ, steps, application ഓൗട്ലൈൻ ഓൗട്ലൈൻ ഓൗട്ലൈൻ. Technical terms English-ഓൗട്ലൈൻ ഓൗട്ലൈൻ.

## Summarize thermodynamic processes. 1 Understanding Interpret enthalpy changes accompanying

Summarize thermodynamic processes. 1 Understanding Interpret enthalpy changes accompanying is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize thermodynamic processes. 1 Understanding Interpret enthalpy changes accompanying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize thermodynamic processes. 1 understanding interpret enthalpy changes accompanying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarize thermodynamic processes. 1 Understanding Interpret enthalpy changes accompanying means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- $\Delta H$  Summarize thermodynamic processes. 1 Understanding Interpret enthalpy changes accompanying  $\Delta H$  definition/meaning  $\Delta H$ .  $\Delta H$   $\Delta H$   $\Delta H$ , steps, application  $\Delta H$   $\Delta H$   $\Delta H$ . Technical terms English- $\Delta H$   $\Delta H$   $\Delta H$ .

## 2 Understanding chemical reactions. Interpret Hess's law of constant heat Applying

2 Understanding chemical reactions. Interpret Hess's law of constant heat Applying is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding chemical reactions. Interpret Hess's law of constant heat Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding chemical reactions. interpret hess's law of constant heat applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding chemical reactions. Interpret Hess's law of constant heat Applying means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഘട്ടം 2 Understanding chemical reactions. Interpret Hess's law of constant heat Applying ഹെസ്സ് നിയമം definition/meaning ഹെസ്സ് നിയമം. ഹെസ്സ് നിയമം ഘട്ടങ്ങൾ, steps, application ഹെസ്സ് നിയമം ഉപയോഗിക്കൽ. Technical terms English-ഘട്ടം ഹെസ്സ് നിയമം.

## 4 summation. Identify the limitations of first law of

4 summation. Identify the limitations of first law of is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 summation. Identify the limitations of first law of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 summation. identify the limitations of first law of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 summation. Identify the limitations of first law of means in the official course context |
| Study lens          | Principle → components or stages → result → application                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-4 summation. Identify the limitations of first law of thermodynamics. definition/meaning. Steps, application. Technical terms English-Malayalam.

## 1 Understanding thermodynamics.

1 Understanding thermodynamics. is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding thermodynamics. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding thermodynamics. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 1 Understanding thermodynamics. means in the official course context         |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-1 Understanding thermodynamics.

താപഗതികശാസ്ത്രം എന്നതിന്റെ definition/meaning എന്താണ്. താപം, താപനില, താപശേഷി, steps, application എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ എഴുതുക.

## Interpret second law of thermodynamics.

Interpret second law of thermodynamics. is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Interpret second law of thermodynamics. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, interpret second law of thermodynamics. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Interpret second law of thermodynamics. means in the official course context |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Interpret second law of thermodynamics.

Interpret second law of thermodynamics. definition/meaning Interpret second law of thermodynamics. Interpret second law of thermodynamics, steps, application Interpret second law of thermodynamics Interpret second law of thermodynamics. Technical terms English- Interpret second law of thermodynamics.

**Interpret third law of thermodynamics. 1 Understanding P-V-T behavior of pure fluids. Processes - Isothermal process - Adiabatic process - Isobaric process - Isochoric process. State Enthalpy changes accompanying chemical reactions: Heat of reaction - Standard heat of reaction - Heat of Formation - Standard heat of formation - Heat of combustion. Hess's law of constant heat summation - Standard heat of reaction from heats of formation. Solve problems on Hess's law of constant heat summation and Standard heat of reaction from heats of formation. Limitations of first law of thermodynamics Second law of thermodynamics - Definition of entropy - Carnot principle - Efficiency of carnot engine - Simple numerical. Third law of thermodynamics. Summarize the applications of thermodynamics in real world systems and**

Interpret third law of thermodynamics. 1 Understanding P-V-T behavior of pure fluids. Processes - Isothermal process - Adiabatic process - Isobaric process - Isochoric process. State Enthalpy changes accompanying chemical reactions: Heat of reaction - Standard heat of reaction - Heat of Formation - Standard heat of formation - Heat of combustion. Hess's law of constant heat summation - Standard heat of reaction from heats of formation. Solve problems on Hess's law of constant heat summation and Standard heat of reaction from heats of formation. Limitations of first law of thermodynamics Second law of thermodynamics - Definition of entropy - Carnot principle - Efficiency of carnot engine - Simple numerical. Third law of thermodynamics. Summarize the applications of thermodynamics in real world systems and is an official topic in Module 2 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Interpret third law of thermodynamics. 1 Understanding P-V-T behavior of pure fluids. Processes - Isothermal process - Adiabatic process - Isobaric process - Isochoric process. State Enthalpy changes accompanying chemical reactions: Heat of reaction - Standard heat of reaction - Heat of Formation - Standard heat of formation - Heat of combustion. Hess's law of constant heat summation - Standard heat of reaction from heats of formation. Solve problems on Hess's law of constant heat summation and Standard heat of

reaction from heats of formation. Limitations of first law of thermodynamics  
 Second law of thermodynamics - Definition of entropy – Carnot principle –  
 Efficiency of carnot engine – Simple numerical. Third law of thermodynamics.  
 Summarize the applications of thermodynamics in real world systems and using  
 standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, interpret third law of thermodynamics. 1 understanding p-v-t behavior of pure fluids. processes - isothermal process - adiabatic process - isobaric process - isochoric process. state enthalpy changes accompanying chemical reactions: heat of reaction - standard heat of reaction - heat of formation - standard heat of formation - heat of combustion. hess's law of constant heat summation - standard heat of reaction from heats of formation. solve problems on hess's law of constant heat summation and standard heat of reaction from heats of formation. limitations of first law of thermodynamics second law of thermodynamics - definition of entropy – carnot principle – efficiency of carnot engine – simple numerical. third law of thermodynamics. summarize the applications of thermodynamics in real world systems and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Interpret third law of thermodynamics. 1 Understanding P-V-T behavior of pure fluids. Processes - Isothermal process - Adiabatic process - Isobaric process - Isochoric process. State Enthalpy changes accompanying chemical reactions: Heat of reaction - Standard heat of reaction - Heat of Formation - Standard heat of formation - Heat of combustion. Hess's law of constant heat summation - Standard heat of reaction from heats of formation. Solve problems on Hess's law of constant heat summation and Standard heat of reaction from heats of formation. Limitations of first law of thermodynamics Second law of thermodynamics - Definition of entropy – Carnot principle – Efficiency of carnot engine – Simple numerical. Third law of thermodynamics. Summarize the applications of thermodynamics in real world systems and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഓർക്കുക Interpret third law of thermodynamics. 1 Understanding P-V-T behavior of pure fluids. Processes - Isothermal process - Adiabatic process - Isobaric process - Isochoric process. State Enthalpy changes accompanying chemical reactions: Heat of reaction - Standard heat of reaction - Heat of Formation - Standard heat of formation - Heat of combustion. Hess's law of constant heat summation - Standard heat of reaction from heats of formation. Solve problems on Hess's law of constant heat summation and Standard heat of reaction from heats of formation. Limitations of first law of thermodynamics Second law of thermodynamics - Definition of entropy - Carnot principle - Efficiency of carnot engine - Simple numerical. Third law of thermodynamics. Summarize the applications of thermodynamics in real world systems and ഓർക്കുക definition/meaning ഓർക്കുക. ഓർക്കുക ഓർക്കുക, steps, application ഓർക്കുക. Technical terms English-ഓർക്കുക.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

thermodynamics.

M2.01 Outline P-V-T behavior of pure fluids 2

Understanding

M2.02 Summarize thermodynamic processes. 1

Understanding

Interpret enthalpy changes accompanying

M2.03 2

Understanding

chemical reactions.

Interpret Hess's law of constant heat

Applying

M2.04 4

summation.

Identify the limitations of first law of

M2.05 1

Understanding

thermodynamics.

M2.06 Interpret second law of thermodynamics. 4

Applying

M2.07 Interpret third law of thermodynamics. 1

Understanding

Series Test – I

1

Contents:

P–V–T behavior of pure fluids.

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 3

## thermodynamic properties of fluids.

This module is presented from the official syllabus scope for Chemical Engineering.  
Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: thermodynamic properties of fluids.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## 2 Understanding thermodynamics

2 Understanding thermodynamics is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding thermodynamics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding thermodynamics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding thermodynamics means in the official course context          |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-2 Understanding thermodynamics

താപഗതികശാസ്ത്രം എന്നതിന്റെ definition/meaning എന്താണ്. താപം, താപനില, താപശേഷി, steps, application എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ എഴുതുക.

## Outline refrigeration cycles and systems

Outline refrigeration cycles and systems is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline refrigeration cycles and systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline refrigeration cycles and systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Definition/identity | What Outline refrigeration cycles and systems means in the official course context |
| Study lens          | Principle → components or stages → result → application                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-ഓൗട്ലൈൻ റിഫ്രിജറേഷൻ സൈക്കിൾ ഓൗട്ലൈൻ

റിഫ്രിജറേഷൻ സൈക്കിൾ റിഫ്രിജറേഷൻ definition/meaning റിഫ്രിജറേഷൻ സൈക്കിൾ. റിഫ്രിജറേഷൻ റിഫ്രിജറേഷൻ, steps, application റിഫ്രിജറേഷൻ റിഫ്രിജറേഷൻ റിഫ്രിജറേഷൻ. Technical terms English-ഓൗട്ട്ലൈൻ റിഫ്രിജറേഷൻ റിഫ്രിജറേഷൻ.

## Outline air standard cycles.

Outline air standard cycles. is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline air standard cycles. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline air standard cycles. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Outline air standard cycles. means in the official course context            |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-ഓൗട്ലൈൻ ഏർവ്വീര്യ സാധാരണ ചക്രങ്ങൾ. ഏർവ്വീര്യ സാധാരണ ചക്രങ്ങൾ  
ഏർവ്വീര്യ definition/meaning ഏർവ്വീര്യ സാധാരണ ചക്രങ്ങൾ. ഏർവ്വീര്യ സാധാരണ ചക്രങ്ങൾ, steps,  
application ഏർവ്വീര്യ സാധാരണ ചക്രങ്ങൾ. Technical terms English-ഓൗട്ലൈൻ  
ഏർവ്വീര്യ സാധാരണ ചക്രങ്ങൾ.

**Summarize thermodynamic properties of fluids 3 Understanding Applications of the laws of thermodynamics -Principle of fluid flow - Nozzles - Mach number - Throttling. Refrigeration - COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants Steam power plant - Rankine cycle - Internal Combustion engines - Otto cycle - Diesel cycle. Thermodynamic properties of fluids - reference properties, energy properties, derived properties - Definition of Helmholtz free energy - Gibbs free energy - Definition of Fugacity - Chemical potential.**

Summarize thermodynamic properties of fluids 3 Understanding Applications of the laws of thermodynamics -Principle of fluid flow - Nozzles - Mach number - Throttling. Refrigeration - COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants Steam power plant - Rankine cycle - Internal Combustion engines - Otto cycle - Diesel cycle. Thermodynamic properties of fluids - reference properties, energy properties, derived properties - Definition of Helmholtz free energy - Gibbs free energy - Definition of Fugacity - Chemical potential. is an official topic in Module 3 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Summarize thermodynamic properties of fluids 3 Understanding Applications of the laws of thermodynamics -Principle of fluid flow - Nozzles - Mach number - Throttling. Refrigeration - COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants Steam power plant - Rankine cycle - Internal Combustion engines - Otto cycle - Diesel cycle. Thermodynamic properties of fluids - reference properties, energy properties, derived properties - Definition of Helmholtz free energy - Gibbs free energy - Definition of Fugacity - Chemical potential. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, summarize thermodynamic properties of fluids 3 understanding applications of the laws of thermodynamics –principle of fluid flow - nozzles – mach number – throttling. refrigeration – cop - carnot cycle - vapour compression cycle - criterion for selection of refrigerants steam power plant – rankine cycle – internal combustion engines – otto cycle - diesel cycle. thermodynamic properties of fluids – reference properties, energy properties, derived properties – definition of helmholtz free energy - gibbs free energy – definition of fugacity – chemical potential. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarize thermodynamic properties of fluids 3 Understanding Applications of the laws of thermodynamics –Principle of fluid flow - Nozzles – Mach number – Throttling. Refrigeration – COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants Steam power plant – Rankine cycle – Internal Combustion engines – Otto cycle - Diesel cycle. Thermodynamic properties of fluids – reference properties, energy properties, derived properties – Definition of Helmholtz free energy - Gibbs free energy – Definition of Fugacity – Chemical potential. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- Summarize thermodynamic properties of fluids 3 Understanding Applications of the laws of thermodynamics –Principle of fluid flow - Nozzles – Mach number – Throttling. Refrigeration – COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants Steam power plant – Rankine cycle – Internal Combustion engines – Otto cycle - Diesel cycle. Thermodynamic properties of fluids – reference properties, energy properties, derived properties – Definition of Helmholtz free energy - Gibbs free energy – Definition of Fugacity – Chemical potential. definition/meaning . steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

thermodynamic properties of fluids.

Summarize the applications of first law of

|               |  |   |
|---------------|--|---|
| M3.01         |  | 2 |
| Understanding |  |   |

thermodynamics

|               |                                          |   |
|---------------|------------------------------------------|---|
| M3.02         | Outline refrigeration cycles and systems | 3 |
| Understanding |                                          |   |

|               |                              |   |
|---------------|------------------------------|---|
| M3.03         | Outline air standard cycles. | 4 |
| Understanding |                              |   |

|               |                                              |   |
|---------------|----------------------------------------------|---|
| M3.04         | Summarize thermodynamic properties of fluids | 3 |
| Understanding |                                              |   |

Contents:

Applications of the laws of thermodynamics –Principle of fluid flow - Nozzles – Mach

number – Throttling.

Refrigeration – COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants

Steam power plant – Rankine cycle – Internal Combustion engines – Otto cycle - Diesel cycle.

Thermodynamic properties of fluids – reference properties, energy properties, derived

properties – Definition of Helmholtz free energy - Gibbs free energy – Definition

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 4

## Outline the fundamental concepts of reaction engineering.

This module is presented from the official syllabus scope for Chemical Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Outline the fundamental concepts of reaction engineering.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Outline chemical reaction engineering

Outline chemical reaction engineering is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline chemical reaction engineering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline chemical reaction engineering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Outline chemical reaction engineering means in the official course context   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഔദ്യോഗിക Outline chemical reaction engineering

ഔദ്യോഗിക definition/meaning ഔദ്യോഗിക. ഔദ്യോഗിക ഔദ്യോഗിക, steps, application ഔദ്യോഗിക ഔദ്യോഗിക. Technical terms English-ഔദ്യോഗിക ഔദ്യോഗിക.

## Summarize the concept of rate of a reaction.

Summarize the concept of rate of a reaction. is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize the concept of rate of a reaction. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize the concept of rate of a reaction. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                        |
|---------------------|----------------------------------------------------------------------------------------|
| Definition/identity | What Summarize the concept of rate of a reaction. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-  
Summarize the concept of rate of a reaction.  
definition/meaning .  
steps, application . Technical terms English-  
.

## Outline the type of reactors.

Outline the type of reactors. is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline the type of reactors. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline the type of reactors. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Outline the type of reactors. means in the official course context           |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Outline the type of reactors.  $\square\square\square\square\square\square\square\square\square\square$   
 $\square\square\square\square$  definition/meaning  $\square\square\square\square\square\square\square\square\square\square$ .  $\square\square\square\square\square$   $\square\square\square\square\square$   $\square\square\square\square\square\square$ , steps,  
 application  $\square\square\square\square$   $\square\square\square\square\square\square\square\square$   $\square\square\square\square\square$ . Technical terms English- $\square$   $\square\square\square\square$   
 $\square\square\square\square\square\square\square\square$ .

**Outline space time and space velocity 1 Understanding Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction - Speed of chemical reaction - Rate equation - State Single and multiple reactions with examples - State elementary and non elementary reactions with examples - State molecularity and order of reaction - State Rate constant & Conversion - State Zero order reaction & First order reaction. Reactors - Principle, diagram and applications - Batch reactor - PFR - MFR Space time - Space velocity. Text / Reference T/R Book Title/Author T1 Chemical Engineering thermodynamic; K V Narayan. R1 Chemical reaction engineering; Octave Levenspiel. Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, R2 Michael M. Abbott. R3 Chemical Engineering Thermodynamics; S. Sundaram. R4 Chemical Reaction Engineering; K. A. Gavhane Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/101/103101004/>**

Outline space time and space velocity 1 Understanding Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction - Speed of chemical reaction - Rate equation - State Single and multiple reactions with examples - State elementary and non elementary reactions with examples - State molecularity and order of reaction - State Rate constant & Conversion - State Zero order reaction & First order reaction. Reactors - Principle, diagram and applications - Batch reactor - PFR - MFR Space time - Space velocity. Text / Reference T/R Book Title/Author T1 Chemical Engineering thermodynamic; K V Narayan. R1 Chemical reaction engineering; Octave Levenspiel. Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, R2 Michael M. Abbott. R3 Chemical Engineering Thermodynamics; S. Sundaram. R4 Chemical Reaction Engineering; K. A. Gavhane Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/101/103101004/> is an official topic in Module 4 of Chemical Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Outline space time and space velocity 1 Understanding Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction - Speed of chemical reaction - Rate equation - State Single and multiple reactions with examples - State elementary and non elementary reactions with examples - State molecularity and order of reaction - State Rate constant & Conversion - State Zero order reaction & First order reaction. Reactors - Principle, diagram and applications - Batch reactor - PFR - MFR Space time - Space velocity. Text / Reference T/R Book Title/Author T1 Chemical Engineering thermodynamic; K V Narayan. R1 Chemical reaction engineering; Octave Levenspiel. Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, R2 Michael M. Abbott. R3 Chemical Engineering Thermodynamics; S. Sundaram. R4 Chemical Reaction Engineering; K. A. Gavhane Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/101/103101004/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline space time and space velocity 1 understanding overview of chemical reaction engineering definition of reaction rate - variables affecting rate of reaction - speed of chemical reaction - rate equation - state single and multiple reactions with examples - state elementary and non elementary reactions with examples - state molecularity and order of reaction - state rate constant & conversion - state zero order reaction & first order reaction. reactors - principle, diagram and applications - batch reactor - pfr - mfr space time - space velocity. text / reference t/r book title/author t1 chemical engineering thermodynamic; k v narayan. r1 chemical reaction engineering; octave levenspiel. introduction to engineering thermodynamics; j.m. smith, hendrick van ness, r2 michael m. abbott. r3 chemical engineering thermodynamics; s. sundaram. r4 chemical reaction engineering; k. a. gavhane online resources sl.no website link 1 <https://nptel.ac.in/courses/103/101/103101004/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Outline space time and space velocity 1 Understanding Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction - Speed of chemical reaction - Rate equation - State Single and multiple reactions with examples - State elementary and non elementary reactions with examples - State molecularity and order of reaction - State Rate constant & Conversion - State Zero order reaction & First order reaction. Reactors - Principle, diagram and applications - Batch reactor - PFR - MFR Space time - Space velocity. Text / Reference T/R Book Title/Author T1 Chemical Engineering thermodynamic; K V Narayan. R1 |

| Aspect        | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Chemical reaction engineering; Octave Levenspiel. Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, R2 Michael M. Abbott. R3 Chemical Engineering Thermodynamics; S. Sundaram. R4 Chemical Reaction Engineering; K. A. Gavhane Online Resources Sl.No Website Link 1 <a href="https://nptel.ac.in/courses/103/101/103101004/">https://nptel.ac.in/courses/103/101/103101004/</a> means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                        |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-ഓൗ Outline space time and space velocity 1 Understanding Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction - Speed of chemical reaction - Rate equation - State Single and multiple reactions with examples - State elementary and non elementary reactions with examples - State molecularity and order of reaction - State Rate constant & Conversion - State Zero order reaction & First order reaction. Reactors - Principle, diagram and applications - Batch reactor - PFR - MFR Space time - Space velocity. Text / Reference T/R Book Title/Author T1 Chemical Engineering thermodynamic; K V Narayan. R1 Chemical reaction engineering; Octave Levenspiel. Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, R2 Michael M. Abbott. R3 Chemical Engineering Thermodynamics; S. Sundaram. R4 Chemical Reaction Engineering; K. A. Gavhane Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/101/103101004/> ഓരോന്നിന്റെയും ഓരോന്നിന്റെ definition/meaning ഓരോന്നിന്റെയും. ഓരോന്നിന്റെ ഓരോന്നിന്റെ steps, application ഓരോന്നിന്റെ ഓരോന്നിന്റെ ഓരോന്നിന്റെ. Technical terms English-ഓ ഓരോന്നിന്റെ ഓരോന്നിന്റെ.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Outline the fundamental concepts of reaction engineering.

|               |                                              |   |
|---------------|----------------------------------------------|---|
| M4.01         | Outline chemical reaction engineering        | 1 |
| Understanding |                                              |   |
| M4.02         | Summarize the concept of rate of a reaction. | 3 |
| Understanding |                                              |   |
| M4.03         | Outline the type of reactors.                | 2 |
| Understanding |                                              |   |
| M4.04         | Outline space time and space velocity        | 1 |
| Understanding |                                              |   |

Series Test – II 1

Contents:

Overview of chemical reaction engineering

Definition of reaction rate - variables affecting rate of reaction – Speed of chemical reaction

– Rate equation – State Single and multiple reactions with examples – State elementary and

non elementary reactions with examples – State molecularity and order of reaction – State

Rate constant & Conversion – State Zero order reaction & First order reaction.

Reactors – Principle, diagram and applications - Batch reactor – PFR – MFR

Space time – Space velocity.

Text / Reference

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## RAPID REFERENCE

## Concept and formula bank

### Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

## RAPID REVISION

### Diagram and source-transcript library

#### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

## OFFICIAL SELF-LEARNING

# Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

## EXAMINATION PREPARATION

### Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

## ACTIVE RECALL

### Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

#### Module 1

**Define or explain Summarize the scope of thermodynamics. 2 Understanding Summarize the properties of a thermodynamic. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize the scope of thermodynamics. 2 Understanding Summarize the properties of a thermodynamic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

### Define or explain 1 Understanding system.. (3 marks)

**Answer:** Begin with the standard definition or identity of *1 Understanding system..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

### Define or explain Analyse state of a system and phase rule. (3 marks)

**Answer:** Begin with the standard definition or identity of *Analyse state of a system and phase rule.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

### Define or explain Interpret Zeroth law of thermodynamics. (3 marks)

**Answer:** Begin with the standard definition or identity of *Interpret Zeroth law of thermodynamics.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## Module 2

### Define or explain Outline P-V-T behavior of pure fluids. (3 marks)

**Answer:** Begin with the standard definition or identity of *Outline P-V-T behavior of pure fluids*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain Summarize thermodynamic processes. 1 Understanding Interpret enthalpy changes accompanying. (3 marks)

**Answer:** Begin with the standard definition or identity of *Summarize thermodynamic processes. 1 Understanding Interpret enthalpy changes accompanying*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain 2 Understanding chemical reactions. Interpret Hess's law of constant heat Applying. (3 marks)

**Answer:** Begin with the standard definition or identity of *2 Understanding chemical reactions. Interpret Hess's law of constant heat Applying*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain 4 summation. Identify the limitations of first law of. (3 marks)

**Answer:** Begin with the standard definition or identity of *4 summation*. Identify the limitations of first law of. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 3

### Define or explain 2 Understanding thermodynamics. (3 marks)

**Answer:** Begin with the standard definition or identity of *2 Understanding thermodynamics*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain Outline refrigeration cycles and systems. (3 marks)

**Answer:** Begin with the standard definition or identity of *Outline refrigeration cycles and systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain Outline air standard cycles.. (3 marks)

**Answer:** Begin with the standard definition or identity of *Outline air standard cycles..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Summarize thermodynamic properties of fluids 3**

**Understanding Applications of the laws of thermodynamics -Principle of fluid flow - Nozzles - Mach number - Throttling. Refrigeration - COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants Steam power plant - Rankine cycle - Internal Combustion engines - Otto cycle - Diesel cycle. Thermodynamic properties of fluids - reference properties, energy properties, derived properties - Definition of Helmholtz free energy - Gibbs free energy - Definition of Fugacity - Chemical potential.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize thermodynamic properties of fluids 3 Understanding Applications of the laws of thermodynamics - Principle of fluid flow - Nozzles - Mach number - Throttling. Refrigeration - COP - Carnot cycle - Vapour compression cycle - Criterion for selection of refrigerants Steam power plant - Rankine cycle - Internal Combustion engines - Otto cycle - Diesel cycle. Thermodynamic properties of fluids - reference properties, energy properties, derived properties - Definition of Helmholtz free energy - Gibbs free energy - Definition of Fugacity - Chemical potential..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 4

**Define or explain Outline chemical reaction engineering. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline chemical reaction engineering.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Summarize the concept of rate of a reaction.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize the concept of rate of a reaction..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Outline the type of reactors.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline the type of reactors..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Outline space time and space velocity 1 Understanding Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction - Speed of chemical reaction - Rate equation - State Single and multiple reactions with examples - State elementary and non elementary reactions with examples - State molecularity and order of reaction - State Rate constant & Conversion - State Zero order reaction & First order reaction. Reactors - Principle, diagram and applications - Batch reactor - PFR - MFR Space time - Space velocity. Text / Reference T/R Book Title/Author T1 Chemical Engineering thermodynamic; K V Narayan. R1 Chemical reaction engineering; Octave Levenspiel. Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, R2 Michael M. Abbott. R3 Chemical Engineering Thermodynamics; S. Sundaram. R4 Chemical Reaction Engineering; K. A. Gavhane Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/101/103101004/>. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline space time and space velocity 1 Understanding Overview of chemical reaction engineering Definition of reaction rate - variables affecting rate of reaction - Speed of chemical reaction - Rate*

equation – State Single and multiple reactions with examples – State elementary and non elementary reactions with examples – State molecularity and order of reaction – State Rate constant & Conversion – State Zero order reaction & First order reaction. Reactors – Principle, diagram and applications - Batch reactor – PFR – MFR Space time – Space velocity. Text / Reference T/R Book Title/Author T1 Chemical Engineering thermodynamic; K V Narayan. R1 Chemical reaction engineering; Octave Levenspiel. Introduction to Engineering Thermodynamics; J.M. Smith, Hendrick Van Ness, R2 Michael M. Abbott. R3 Chemical Engineering Thermodynamics; S. Sundaram. R4 Chemical Reaction Engineering; K. A. Gavhane Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/101/103101004/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **Summarize the scope of thermodynamics.**

**2 Understanding Summarize the properties of a thermodynamic.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **Outline P-V-T behavior of pure fluids.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **2 Understanding thermodynamics**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 4

1-mark definition: State the meaning of **Outline chemical reaction engineering**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

### Exam checklist

#### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

#### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.

- Separate official syllabus information from personal elaboration.

## REFERENCE VOCABULARY

# Glossary

### Study glossary

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

# References and web resources

### References listed in the official PDF

References are included in the official source PDF.

### Web resources listed in the official PDF

- Sl.No Website Link
- <https://nptel.ac.in/courses/103/101/103101004/>

## IN-PAGE SEARCH

# Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3071. Verify institutional updates against the current official curriculum.